



哈爾濱工業大學 (深圳)
HARBIN INSTITUTE OF TECHNOLOGY

实验报告

开课学期: 2024 秋季
课程名称: 数据库系统
实验名称: 数据库系统功能实现
实验性质: 设计型
实验学时: 8 地点: T2210
学生班级: 5
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实验与创新实践教育中心制

2024 年 9 月

1 实验环境

- 操作系统: Linux
- 编程语言: C++
- 编译环境: g++
- 依赖库: Linux 标准库

2 测试结果

请截图测试结果

2.1 任务 1.1 磁盘存储管理器测试截图

```
root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# make disk_manager_test
[ 20%] Built target gtest
[ 40%] Built target gtest_main
Consolidate compiler generated dependencies of target storage
[ 50%] Building CXX object src/storage/CMakeFiles/storage.dir/buffer_pool_manager.cpp.o
[ 60%] Linking CXX static library ../../lib/libstorage.a
[ 80%] Built target storage
[ 90%] Linking CXX executable ../../bin/disk_manager_test
[100%] Built target disk_manager_test
root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# ./bin/disk_manager_test
Running main() from /root/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc
[=====] Running 2 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 2 tests from DiskManagerTest
[ RUN      ] DiskManagerTest.FileOperation
[      OK  ] DiskManagerTest.FileOperation (8 ms)
[ RUN      ] DiskManagerTest.PageOperation
[      OK  ] DiskManagerTest.PageOperation (15 ms)
[-----] 2 tests from DiskManagerTest (24 ms total)

[-----] Global test environment tear-down
[=====] 2 tests from 1 test suite ran. (24 ms total)
[ PASSED  ] 2 tests.
```

2.2 任务 1.2 缓冲池替换策略测试截图

```
● root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# make lru_replacer_test
[ 25%] Built target gtest
[ 50%] Built target gtest_main
Consolidate compiler generated dependencies of target lru_replacer
[ 75%] Built target lru_replacer
[100%] Built target lru_replacer_test
● root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# ./bin/lru_replacer_test
Running main() from /root/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc
[=====] Running 3 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 3 tests from LRUReplacerTest
[ RUN      ] LRUReplacerTest.SimpleTest
[       OK ] LRUReplacerTest.SimpleTest (0 ms)
[ RUN      ] LRUReplacerTest.MixTest
[       OK ] LRUReplacerTest.MixTest (17 ms)
[ RUN      ] LRUReplacerTest.ConcurrencyTest
[       OK ] LRUReplacerTest.ConcurrencyTest (123 ms)
[-----] 3 tests from LRUReplacerTest (141 ms total)

[-----] Global test environment tear-down
[=====] 3 tests from 1 test suite ran. (141 ms total)
[ PASSED  ] 3 tests.
```

2.3 任务 1.3 缓冲池管理器测试截图

```
● root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# make buffer_pool_manager_test
[ 20%] Built target gtest
[ 40%] Built target gtest_main
Consolidate compiler generated dependencies of target storage
[ 80%] Built target storage
[ 90%] Linking CXX executable ../../bin/buffer_pool_manager_test
[100%] Built target buffer_pool_manager_test
● root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# ./bin/buffer_pool_manager_test
Running main() from /root/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc
[=====] Running 4 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 4 tests from BufferPoolManagerTest
[ RUN      ] BufferPoolManagerTest.SimpleTest
[       OK ] BufferPoolManagerTest.SimpleTest (28 ms)
[ RUN      ] BufferPoolManagerTest.LargeScaleTest
[       OK ] BufferPoolManagerTest.LargeScaleTest (194 ms)
[ RUN      ] BufferPoolManagerTest.MultipleFilesTest
[       OK ] BufferPoolManagerTest.MultipleFilesTest (1537 ms)
[ RUN      ] BufferPoolManagerTest.ConcurrencyTest
[       OK ] BufferPoolManagerTest.ConcurrencyTest (637 ms)
[-----] 4 tests from BufferPoolManagerTest (2398 ms total)

[-----] Global test environment tear-down
[=====] 4 tests from 1 test suite ran. (2398 ms total)
[ PASSED  ] 4 tests.
```

2.4 任务 2 记录管理器测试截图

```
● root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# make record_manager_test
[ 8%] Built target gtest
[ 16%] Built target gtest_main
[ 33%] Built target storage
[ 45%] Built target index
[ 58%] Built target recovery
[ 70%] Built target transaction
[ 79%] Built target system
Consolidate compiler generated dependencies of target record
[ 91%] Built target record
[100%] Built target record_manager_test
● root@iZwz91cn4x0do10eczmhc0Z:~/rucbase-lab/build# ./bin/record_manager_test
Running main() from /root/rucbase-lab/deps/googletest/googletest/src/gtest_main.cc
[=====] Running 2 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 2 tests from RecordManagerTest
[ RUN      ] RecordManagerTest.SimpleTest
insert 455
delete 278
update 267
[ OK      ] RecordManagerTest.SimpleTest (572 ms)
[ RUN      ] RecordManagerTest.MultipleFilesTest
[ OK      ] RecordManagerTest.MultipleFilesTest (274 ms)
[-----] 2 tests from RecordManagerTest (847 ms total)

[-----] Global test environment tear-down
[=====] 2 tests from 1 test suite ran. (847 ms total)
[ PASSED  ] 2 tests.
```